

NYC Open Data Automated Dataset Description Generation using Large Language Models

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1 Introduction

NYC Open Data is a public data repository launched by the City of New York that provides thousands of datasets covering many domains such as transportation, housing, public health, education and public safety. Despite the scale of the repository, the quality of dataset descriptions is often insufficient: many descriptions fail to convey what the dataset actually contains, what its columns mean, or how it should be interpreted. Given that the repository’s search functionality relies heavily on complete descriptions, missing or shallow descriptions make relevant datasets hard to discover.

Recent work on Automated Dataset Description Generation (AutoDDG) has shown that LLMs can generate descriptions from dataset profiling outputs, offering a promising path toward improving documentation at scale [1]. In the context of NYC Open Data, AutoDDG descriptions can be augmented with contextual information like column level dictionaries, agency attribution, update frequency, tags, and notes to improve the accuracy and completeness of descriptions. This is particularly useful to data scientists, who spend roughly 80% of their time preparing data for analysis [2], because better descriptions allow for more efficient data search and discovery. Beyond description generation, we also aim to help users determine whether a dataset can actually answer their questions.

The existing AutoDDG approach has limitations that hinders its use for NYC Open Data. First, users must manually download datasets and input function parameters before generating descriptions. These steps are impractical during dataset discovery, when users are searching for datasets to use. Also, the underlying LLM cannot validate inputs, causing any input errors to affect the generated description. For example, passing incorrect date ranges into the model will cause final descriptions to feature incorrect dates.

This project aims to tailor AutoDDG to NYC Open Data through two main avenues. Firstly, by redesigning the input to be a single dataset URL, which eliminates the need for local downloads. Secondly, to enable descriptions to be more complete, by incorporating contextual content from the repository. Through these enhancements, we aim to utilize AutoDDG to improve data discovery and understanding on NYC Open Data.

2 Problem Formulation

Given an NYC Open Data URL, we would like to automatically generate dataset descriptions that are comprehensive, accurate, and readable for increased dataset discoverability. These summaries should entail the following properties of the dataset: column names, data types, data constraints and potential use cases. We address this through the following sub problems:

Extracting Dataset and Metadata. Designing a function that takes an NYC Open Data URL and outputs the dataset, title, existing description (if any), data dictionary and other relevant contextual

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information to feed into AutoDDG. This is non-trivial as dataset publishers use inconsistent formatting of the data dictionary which makes discovery and extraction difficult for certain edge cases.

Description generation. Supplementing existing prompts in AutoDDG and crafting new prompts to incorporate additional context found in NYC Open Data. We modify the most impactful areas in AutoDDG to incorporate additional context for richer and more complete descriptions.

Constraint generation. Crafting prompts and designing functions that output column constraints given sample data and a data dictionary. These can be used by the user as an initial screening for dataset suitability.

Dataset question evaluator. Crafting prompts and a function that takes in the generated dataset description, the dataset’s properties (semantic profile, etc) and a question by the user, and returns a comprehensive but concise answer to the question.

3 Related Work

NYC Open Data. Legislation has mandated that NYC public datasets and their associated dictionaries be made available on NYC Open Data [3]. Despite this, there are still instances of inadequate or misleading descriptions [1]. Therefore, we conducted a preliminary analysis on NYC Open Data to check the quality of the metadata on the repository. We found that roughly 85% of education datasets contain data dictionaries. Furthermore, out of 200 randomly sampled education datasets, approximately 60% of descriptions were either missing or consisted only of the dataset title. Overall, our analyses identify a significant gap in description quality on NYC Open Data.

Dataset Documentation. The original motivation for AutoDDG was that datasets become undiscoverable due to missing or inadequate descriptions [1]. Even when datasets are discoverable, we have found that inadequate dataset documentation still presents issues. By lacking dataset documentation, data practitioners spend significant time inspecting raw data to recover context that should have been documented in the first place [4]. Proper descriptions can also assist practitioners on downstream tasks like feature engineering and data cleaning [4]. As such, we were motivated to extend AutoDDG’s capabilities to highlight data constraints such as allowed values, formats, requirements, and any miscellaneous notes.

AutoDDG. AutoDDG receives a dataset title, existing description and data sample and generates a new dataset description. It decomposes inputs into a content profiler, a semantic profiler, and a topic generator that ultimately creates a prompt for an LLM, which then generates a dataset description and a search-focused dataset description. While AutoDDG has improved dataset search capabilities, the approach still has its limitations. The reliance on data samples can create misleading descriptions that fail to capture crucial global properties of the data [1]. Furthermore, characteristics like datatypes and column semantics are inferred from the sample, which can be incorrect [1]. Finally, during dataset discovery, the requirement of manually downloading datasets and function inputs is also tedious and storage-consuming when screening for datasets to use. Therefore, our approach aims to address the aforementioned limitations within the NYC Open Data environment.

4 Method

Extracting Dataset and Metadata. We modified AutoDDG’s workflow (Fig. 5) to receive a NYC Open Data URL as input and generate descriptions. This choice was made to remove the requirement for users to download the dataset and manually supply contextual information. This modification consists of a small set of functions that access the NYC Open Data API through the URL and return all information required for AutoDDG, such as the title, description, dataset, category and column

types. The design of these functions is based on the structure of NYC Open Data URLs:

`https :// data.cityofnewyork.us / Education / 2012-SAT-Results / f9bf-2cp4 /`
 protocol host category title identifier

Each URL contains two main components for endpoint access: API base (host) and the dataset identifier. These two values fetch structured metadata from the Socrata Open Data API and content from the dataset’s rendered web page. In addition, the data dictionary and dataset are retrieved with the same URL but separately from metadata. The dictionary endpoint returns a list of all files attached to the dataset, which may include unrelated documents. Thus, we must inspect each attachment to identify the data dictionary. In terms of the dataset, the dataset endpoint has a size limitation of 50000 rows per request, meaning that larger datasets must be retrieved iteratively.

All information obtained from NYC Open Data is then loaded directly into Pandas DataFrames through download links, eliminating the need for any local downloads.

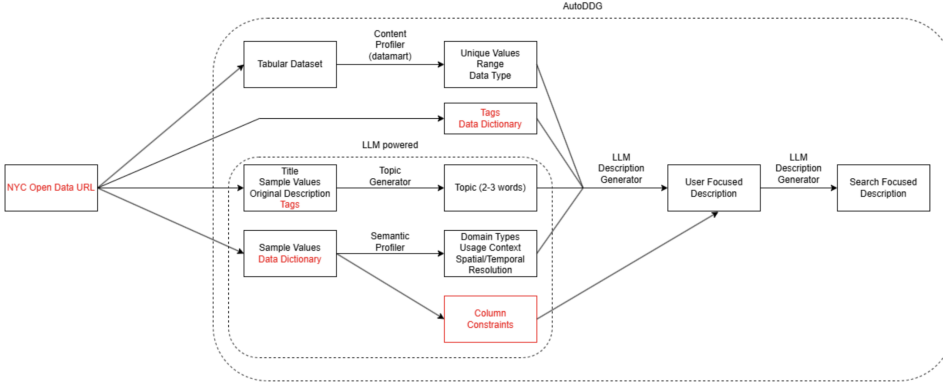


Fig. 1. Overall AutoDDG Workflow and Modifications.

Data Dictionaries. As mentioned above, we were required to search through all dataset attachments to find data dictionaries. These were not always explicitly labelled as data dictionaries and were stored in different file formats (xls, xlsx, csv).

During identification, we iterated through csv files and each sheet in excel files for any given dataset. We assumed most data dictionaries would follow the NYC Open Data data dictionary template [5], where dataset column names primarily appeared in the first column. So, we checked each excel sheet’s first three columns against the dataset’s column names to account for any mild formatting variation.

Given that naming was inconsistent in some instances, we had to normalize data dictionary column names before conducting similarity scoring. This included dropping text in parentheses, expanding symbols (% → percent, etc.), and expanding abbreviations (qty→quantity, etc). After which, we conducted a lexical similarity test (rapidfuzz) to ascertain whether we had a match. If we did not meet our threshold for any sheet, we fell back to semantic similarity (fastembed) as a precaution for users who paraphrased the column names in the data dictionary.

As for processing, we parsed data dictionaries into python dictionaries with column names as keys and sub-dictionaries of metadata as values. This data structure was chosen to capture each

dataset column and all data associated with it in the data dictionary. An example of this on a sample dataset is exemplified below:

```
{
  'dca_license_number':
    {'column_description': 'An identification number issued to businesses/individuals to operate legally
    for the duration of their license term or until the license is revoked.'},
  'Business_name':
    {'column_description': 'The legal business name as filed with the New York State Secretary of
    State or County Clerk or, if an Individual license, the individual's first name and last name.'},
  'business_name_2':
    {'column_description': 'If applicable, the Doing-Business-As (DBA) or trade name.'},
  'industry':
    {'column_description': 'The category of business activity that requires a DCA license.'},
  'event_type':
    {'column_description': 'Indicates if the license was suspended, revoked, surrendered, or
    reinstated.'},
    'additional_notes': 'Suspended\nRevoked\nSurrendered\nReinstated'},
  'event_date':
    {'column_description': 'Date when the license was suspended, revoked, surrendered, or
    reinstated.'},
  'status':
    {'column_description': 'The status of the license as of the data extract date.',
    'additional_notes': 'Active\nInactive'}
}
```

Fig. 2. Data Dictionary Sample.

Description Generation. To enhance the generated dataset descriptions by the original AutoDDG, we integrated additional context into the prompts in the form of a data dictionary and tags into the prompt pipeline. In Topic Generator, we included tags for more accurate topic generation, as they specify the domain and related keywords. The data dictionary serves as a critical source of feature-level metadata, providing column descriptions and value constraints. By integrating this into the Semantic Analyzer, we were able to enhance the model’s ability to map raw schema to specific domain types and infer nuanced usage contexts. While testing this initial configuration, we discovered that a lot of additional context was diluted in the Semantic Analyzer and did not make a proportional change to the final description. Therefore, we chose to bypass the analyzer’s bottleneck by also injecting the data dictionary and tags directly into the final generation prompt. This iteration gave us a fuller description at first glance, but seemed to be missing vital information present in the original description. So, we increased the word count of the User-Focused description from 100 to 150 words to accommodate extra context without sacrificing clarity. We also added a section in the Search-Focused description that entails some questions that this dataset can potentially tackle. This was to generate more key words for better search visibility.

Constraint Generation. Most data dictionaries have a column for expected values where it specifies the range and possible inputs of a field. Therefore, we crafted a prompt that gave the LLM the data dictionary to output constraints for relevant columns. In order to not dilute vital information, we removed column description from the data dictionary before feeding it into the prompt. Furthermore, since it is possible that the information contained in data dictionaries are inaccurate, we provided some sample values to the LLM to verify and refine the constraints. The constraints were then appended to our User-Focused Description.

Question Evaluation. Prior work shows that LLMs can effectively use metadata and descriptions to determine the relevance of a dataset to a user’s questions [6]. Thus, we enabled AutoDDG to assess dataset suitability. We created a function that takes two inputs: a user question and a context object containing attributes like the generated description, dataset topic, profiled metadata and

a sample of records. The function then constructs a prompt instructing the LLM to determine whether the dataset can answer the user question and justify its assessment. Finally, the output consists of a “yes” or “no” and a justification for the response.

5 Evaluation

We use three strategies to evaluate our approach and its capabilities, all of which are compared against AutoDDG as a baseline.

1. Evaluate completeness, conciseness, and readability of our descriptions.
2. Evaluate the ability of our descriptions to answer questions.
3. Evaluate the search effectiveness of our descriptions.

Pairwise Comparisons. As noted when evaluating the original AutoDDG, pairwise LLM evaluations have been confirmed to align approximately with human judgments in large scale studies [1]. We extended this evaluation strategy to capture whether, in the context of NYC Open Data, the new AutoDDG approach was able to generate better descriptions based on completeness, conciseness, and readability. Naturally, we acknowledged that the constraints added to our User-Focused Descriptions have a trade off between completeness and conciseness, but we aimed to evaluate description quality overall.

For 572 datasets across the education, business, and healthcare categories within NYC Open Data, we generated descriptions using the original AutoDDG and the new framework. We then performed pairwise evaluations between the two for each dataset. In the output, the LLM determines the winner and provides a justification. For a more controlled experiment, we adjusted both versions of AutoDDG to output 150-word descriptions, as varying lengths tend to heavily impact LLM evaluation.

Category	Total Datasets	Original Wins	New Approach Wins	Win Rate
Business	61	6	55	90%
Healthcare	65	13	52	80%
Education	446	70	376	84%
Total	572	89	483	84%

Fig. 3. Table of Original AutoDDG wins against New Approach wins for each category.

Judging by win rates, we see that our approach produced better descriptions 84% of the time. The reasons given by the LLM for choosing one dataset over another were generally homogenous, citing a more comprehensive overview while maintaining readability and conciseness. The areas where the baseline wins are usually those where no data dictionary is provided or where additional information has caused the description to focus on specific unimportant details.

Question Comparisons. To evaluate the ability of our approach to answer questions, we compare both descriptions against a fixed set of questions commonly asked when assessing a dataset. We define four evaluation questions, each targeting a dimension of dataset understanding:

- Limitations and data quality: What are some limitations or data quality issues?
- Row semantics: What does each row represent?
- Temporal range: What is the temporal range of data?
- Joinability: What identifiers can be used to join with other NYC datasets?

Similarly to the previous evaluation, we generate descriptions using both the original and new AutoDDG frameworks. Then, the LLM determines if a given description can answer the four questions listed above. A question is considered answered if the description contains the information to be able to answer it. The score for each question is the percentage of datasets that were able to answer it, allowing a direct comparison of how much more useful information the new approach conveys relative to the original AutoDDG. The results show an overall improvement relative to the original across all questions. Questions 1 and 4 are answered significantly better by the new approach due to the additional context of data dictionaries. However, questions 2 and 3 show no significant difference between two approaches. Overall, the new method achieves higher scores for questions 1 and 4, indicating improved effectiveness.

Question	Approach	Education Answer Rate	Business Answer Rate	Health Answer Rate	Total Answer Rate
What are some limitations or data quality issues?	Original	2.46%	4.92%	4.69%	2.97%
	New	11.19%	22.95%	20.31%	13.46%
What does each row represent?	Original	97.99%	100.00%	95.31%	97.90%
	New	99.78%	100.00%	96.88%	99.48%
What is the temporal range of data?	Original	25.95%	70.49%	76.56%	36.36%
	New	28.19%	72.13%	71.88%	37.76%
What identifiers can be used to join with other NYC datasets?	Original	46.98%	39.34%	17.19%	42.83%
	New	60.85%	55.74%	29.69%	56.82%

Fig. 4. Percentage of Datasets Answering Each Question by Category.

Visibility of Search-Focused Descriptions. To evaluate the effectiveness of our Search-Focused Descriptions, we compared retrieval quality between our descriptions and the original AutoDDG descriptions. For each dataset, we used an LLM to generate three synthetic analytical search queries grounded in the dataset’s data dictionary and User-Focused Descriptions. We chose to use three search queries to reduce variance in the retrieval metric, as one misphrased query could tank that dataset’s contribution to the metric. Each query was also generated differently to capture different kinds of search behavior. The first query is one that mentions a field explicitly, such as “avg SAT math score by school”. The second query adds onto this with a field filter, such as “test scores by borough 2019”. Our last query is a free-form analytical question, such as “How does the distribution of math scores vary by school across NYC?”. These queries are phrased in the prompt to be highly specific to the dataset. This is important as each query’s source dataset is treated as the single correct answer in our evaluation, and if a query is too generic, the system would unfairly penalize a “wrong” dataset that was actually a reasonable hit.

We then indexed each Search-Focused Description separately into Elasticsearch using its default BM25 scoring and ran every query against both indexes. We chose Elasticsearch because it is the engine NYC OpenData uses, which makes our results a closer proxy for how the descriptions would perform for actual users. We measured Recall@1, Recall@5, Recall@10, MRR, and NDCG@10 to capture both top-rank accuracy and overall ranking quality. We ran this on 572 datasets stretching

Retrieval Metric	Original Approach	New Approach	delta
Recall@1	0.3741	0.4749	+0.1008
Recall@5	0.5822	0.6702	+0.0880
Recall@10	0.6550	0.7325	+0.0775
MRR	0.4623	0.5586	+0.0963
NDCG@10	0.5086	0.6005	+0.0919

Fig. 5. Table of Original AutoDDG against New Approach on Search-Focused Descriptions Performance.

across three domains (education, business, and health) and recorded the overall weighted scores below.

The search-focused descriptions outperformed the original AutoDDG descriptions on every retrieval metric measured. The largest gain was on Recall@1 with a 27% relative improvement, showing that the new descriptions are effective at surfacing the correct dataset at the top of the result list. Improvements were consistent across binary recall, mean reciprocal rank, and NDCG, which rules out metric-specific artifacts. Overall across metrics, our approach seems to have improved search performance by about 10%.

6 Conclusion and Future Work

NYC Open Data legally mandates city agencies to publish datasets with data dictionaries and additional context. Despite this, we found that up to 60% of datasets in a sample show inadequate descriptions, hindering search capability and efficient data understanding. This paper tailors the AutoDDG framework to the repository specifically, by incorporating data dictionaries and tags into the semantic profiler and topic generator. In doing so, we enabled the LLM to make better inferences about column relationships and dataset usage, and raised constraints for easier data understanding. Our evaluations support this, showing the tailored approach winning 84% of pairwise evaluations, answering more questions, and improving search capabilities by 10% in comparison to the original AutoDDG framework.

Limitations and Future Directions. LLMs operate within a fixed input context window which restricts the amount of information that can be passed to the model [7]. Therefore, the information we pass into the prompt can exceed the token limit and stop the model. One area of extension is to develop methods for filtering the most relevant content from the metadata and data sample, in order to reduce the input size while maintaining high quality descriptions. Furthermore, LLM hallucinations continue to be a limitation to our approach. Detection and mitigation strategies to handle hallucinations and provide complete and accurate descriptions remains a significant topic of research [1]. While most data dictionaries in NYC Open Data are accurate, we observed many that contained incomplete or incorrect information. Fed into our new AutoDDG framework, the incorrect information will likely produce uninformative or misleading descriptions. This is an inevitable part of using metadata that originates from many different sources with no fixed structure or validation.

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